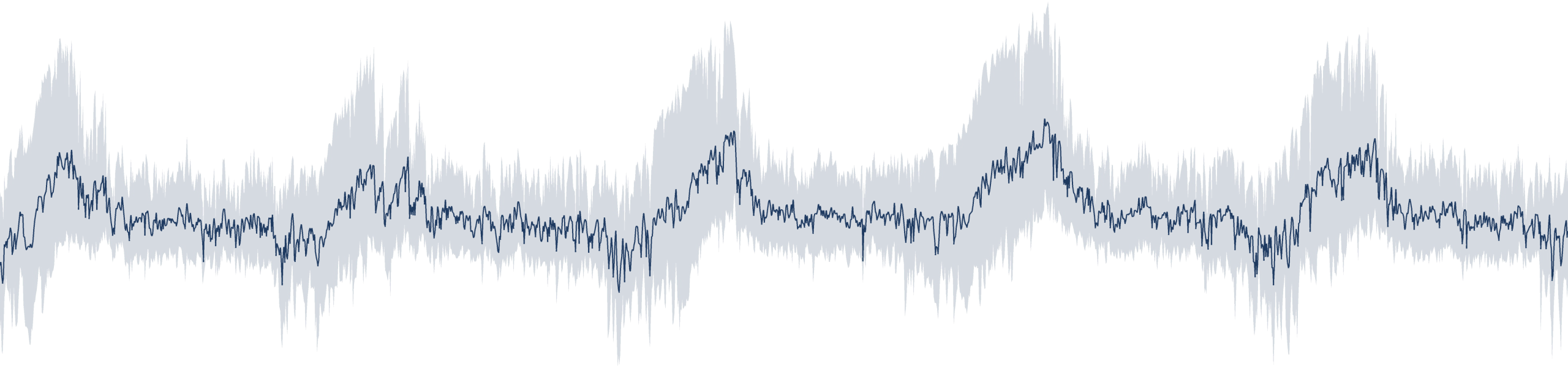
CE436 · Discrete-time signal processing · Final defense

Weather forecasting on FPGA

Parallel fixed-point inference and verification on a Cyclone II DE2 board



Every line below is real data: 43,824 hourly temperature observations, Ho Chi Minh City, 2021–2025 (NASA POWER)

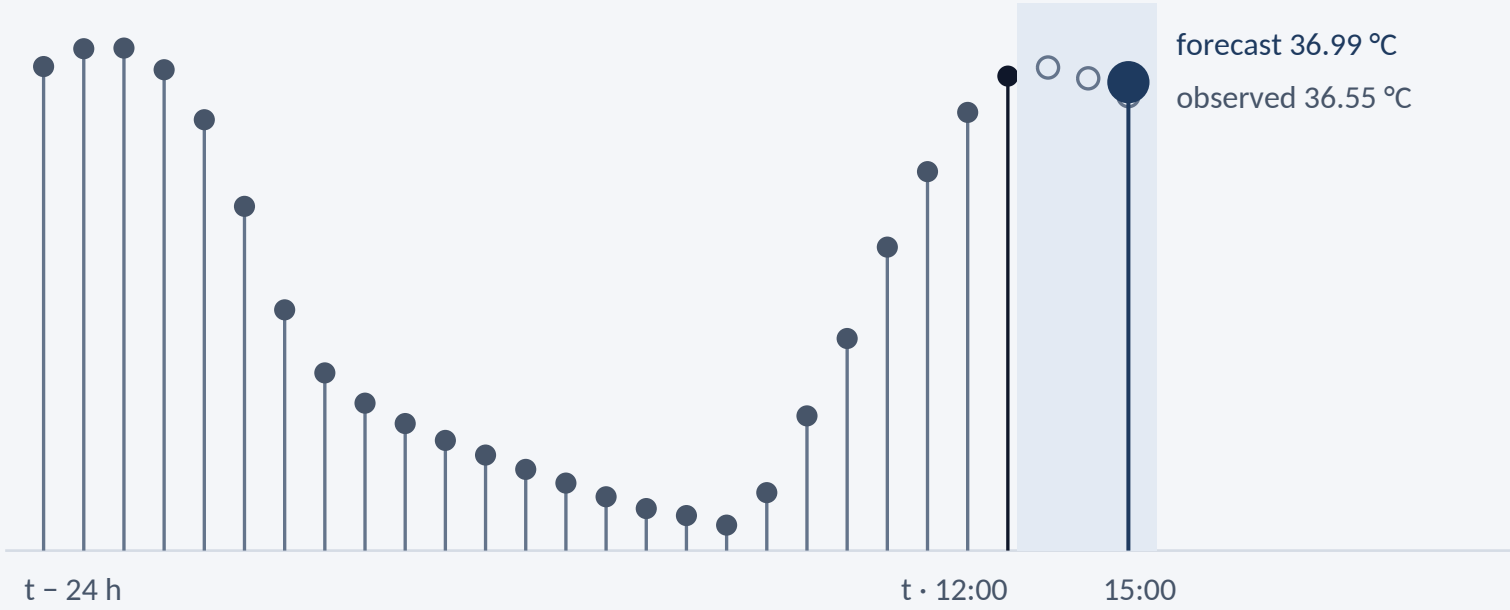
Forecast the weather three hours ahead

+3

hours ahead

Five quantities from each hourly frame

- Temperature
- Pressure
- Humidity
- Wind speed
- Wind direction



Real 2025 test-set window (2025-03-24, local time) · temperature lane of the deployed model



From one temperature output to five weather quantities

1 forecast quantity

Temperature prototype

Model	Three-parameter seasonal OLS
Arithmetic	One lane · 16-bit Q8
Direction	No direction path

5 weather quantities

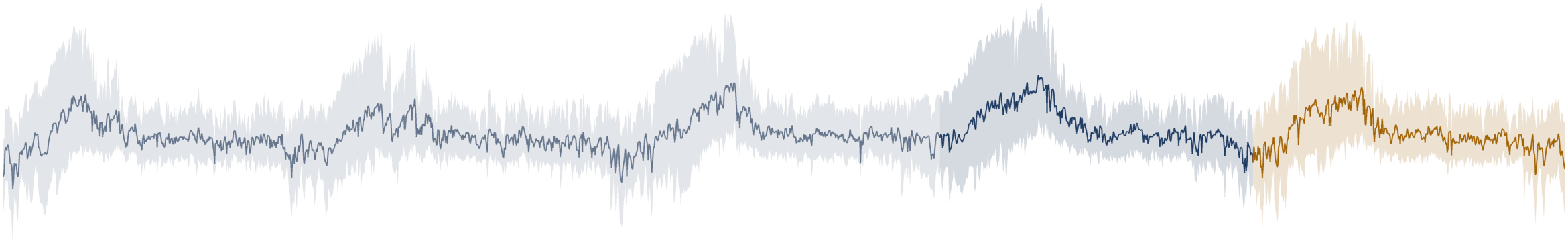
Current weather design

Model	Seven-lag regression per channel
Arithmetic	Six lanes · 18-bit Q8
Direction	Cosine + sine, CORDIC angle

Unchanged: hourly inputs and the three-hour target horizon

03 Data

A chronological split: the model never sees the future



2021–2023

26,253

Training

Fits the coefficients

2024

8,781

Validation

Selects the model

2025

8,757

Test

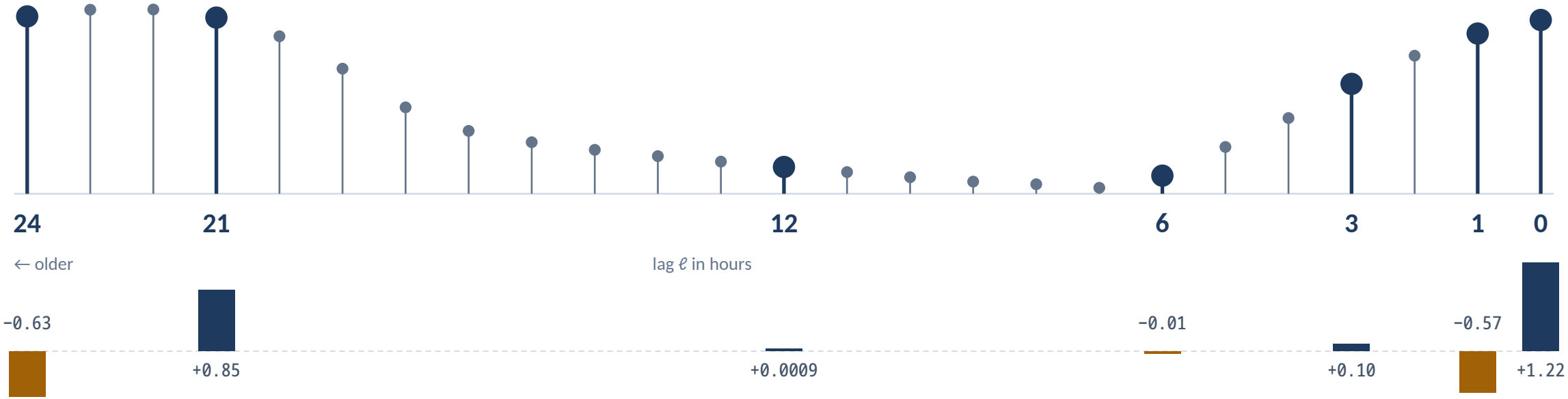
Held out, frozen model

Usable examples: each needs a valid 25-hour history and a target three hours later. NASA POWER hourly, Ho Chi Minh City, UTC.

Seven lags per channel, one linear sum

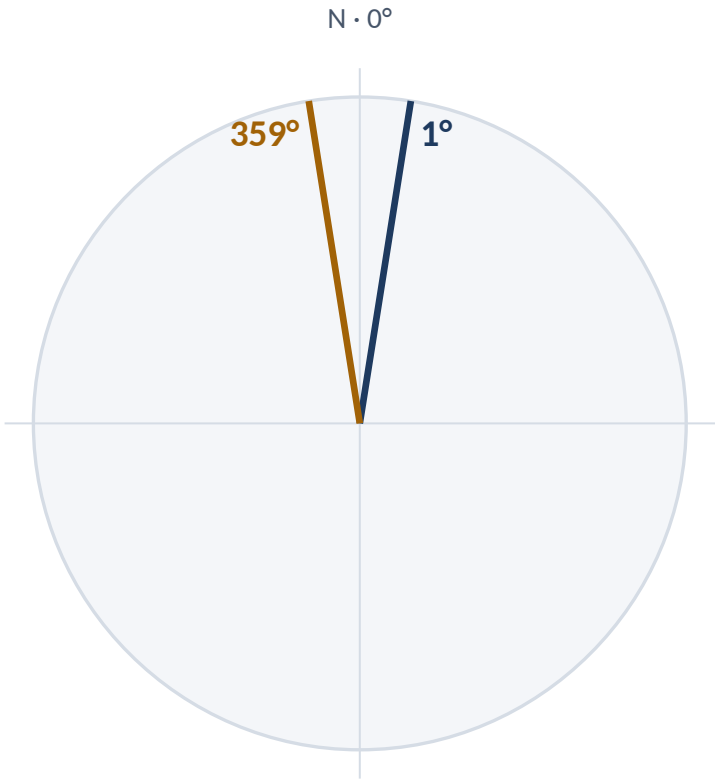
$$\hat{x}(t+3) = b + \sum w_k \cdot x(t - \ell_k)$$

k = 1 … 7 · each channel uses only its own history



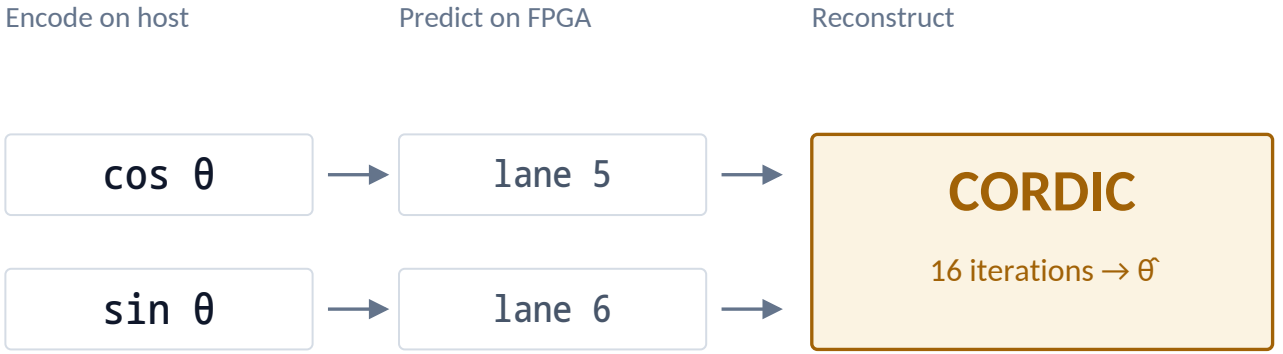
Deployed temperature weights: the last hours and the same hours yesterday dominate

Wind direction wraps around at 0°



2° apart, not 358°

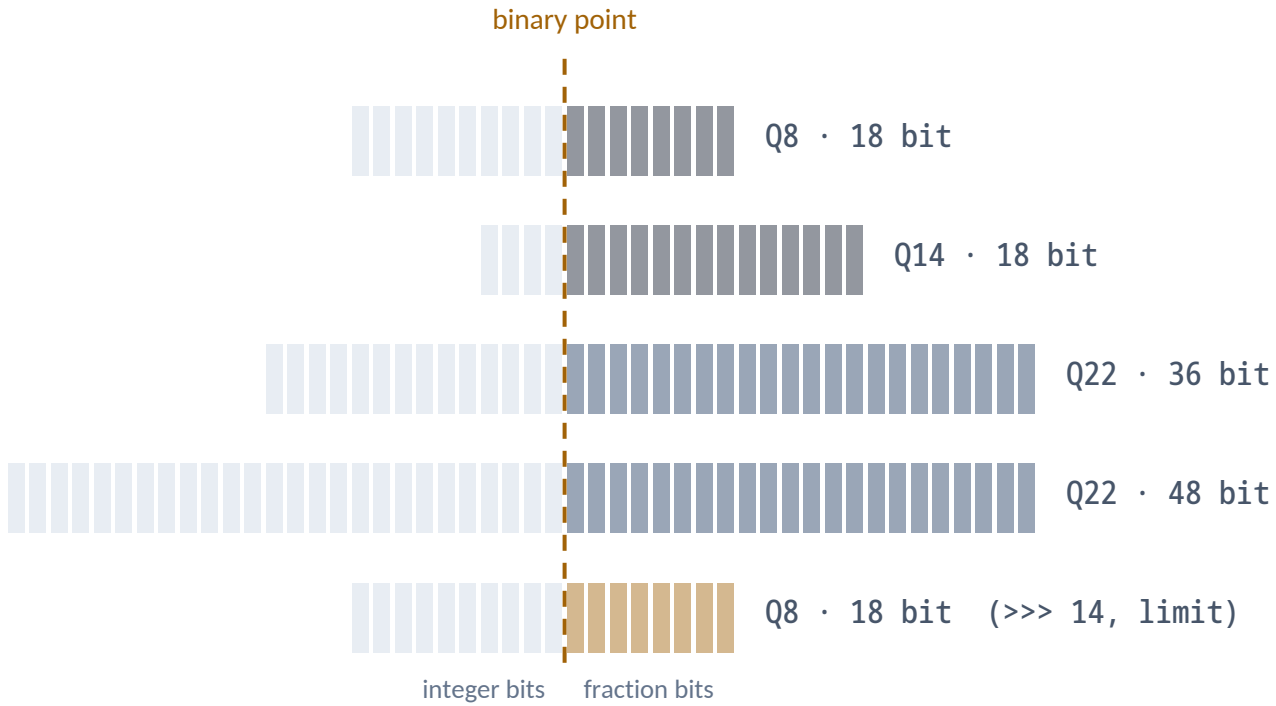
Rays drawn wider apart than 2° for legibility



Each angle becomes two smooth components, so nearby directions stay numerically close. The FPGA predicts both and turns them back into one angle.

Zero component pair → direction flagged undefined

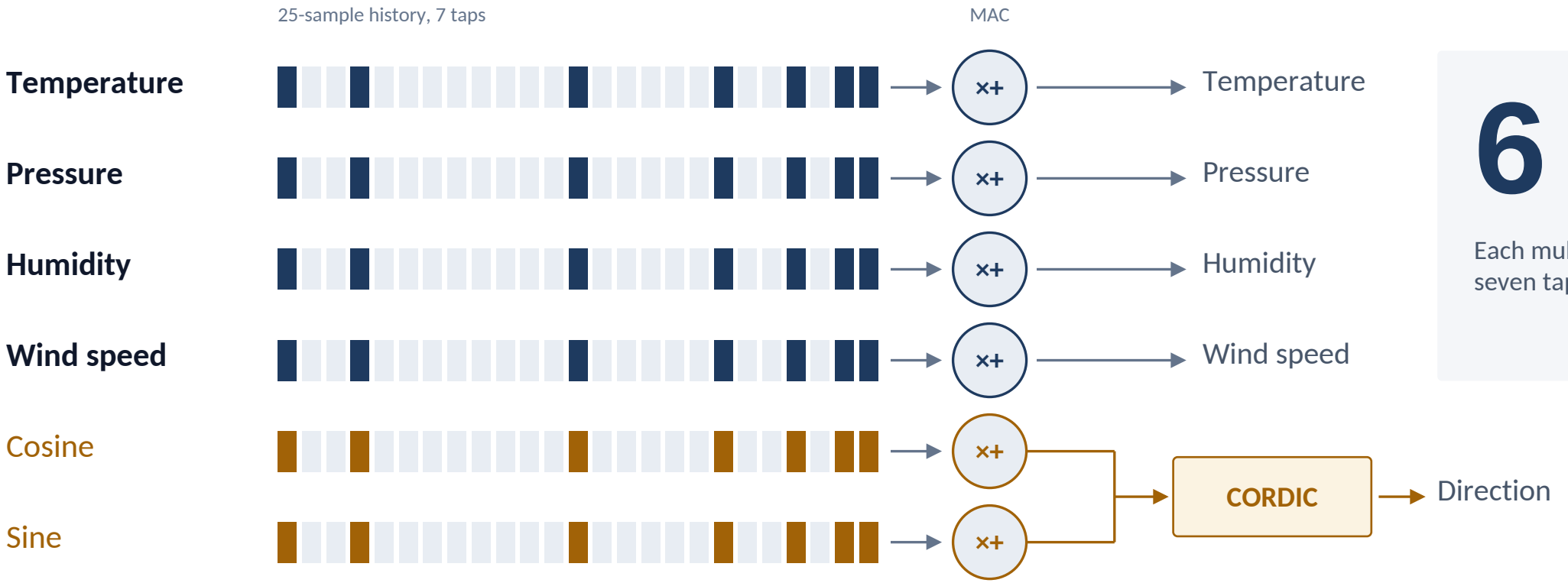
Fixed point: keep every bit until the final shift



$30\ °C \times 2^{8} = \mathbf{7680}\ \text{raw}$

Aligned bias, floor scaling, range limits. The C++ reference uses identical rules, so any mismatch is an implementation bug, not forecast error.

Six lanes in parallel, one multiplier each



6 **multipliers, not 42**

Each multiplier walks its seven taps over time.

History advances only when a frame is accepted. The wrapper aligns scalar results with the reconstructed direction.

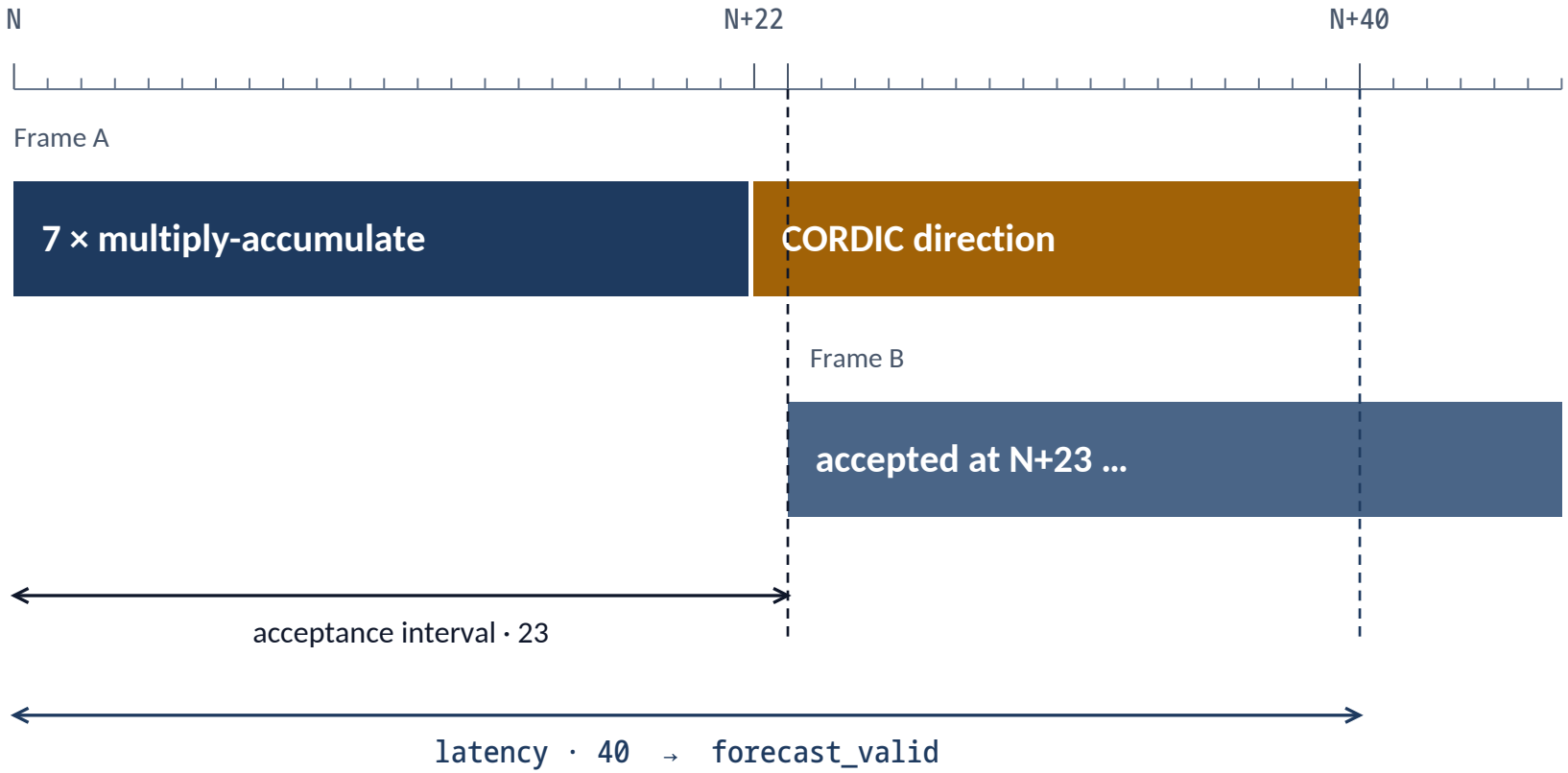
40 cycles of latency, a new frame every 23

40

cycles of latency

0.80 µs at 50 MHz

First forecast follows the 25th accepted frame: 24 frames fill the history.



An idle clock is not another hour: history shifts only on accepted frames.

Three independent checks, each closer to silicon

Stage 1

RTL + DPI-C

1,124

forecasts checked

from 1,200 accepted frames

C++ golden model, cycle-level scoreboard

Stage 2

Core SDF

264

forecasts checked

from 288 input frames

Gate-level netlist, max delays, 20 ns clock

Stage 3

Physical JTAG

104

forecasts checked

from 128 input frames

Real responses read from the DE2 board

Checked at every stage

- Values
- Transaction order
- Reset behavior
- Cycle timing

The model beats both baselines on every quantity

Temperature
MAE · °C

−39%

vs best baseline

Pressure
MAE · kPa

−60%

vs best baseline

Humidity
MAE · pp

−31%

vs best baseline

Wind speed
MAE · m/s

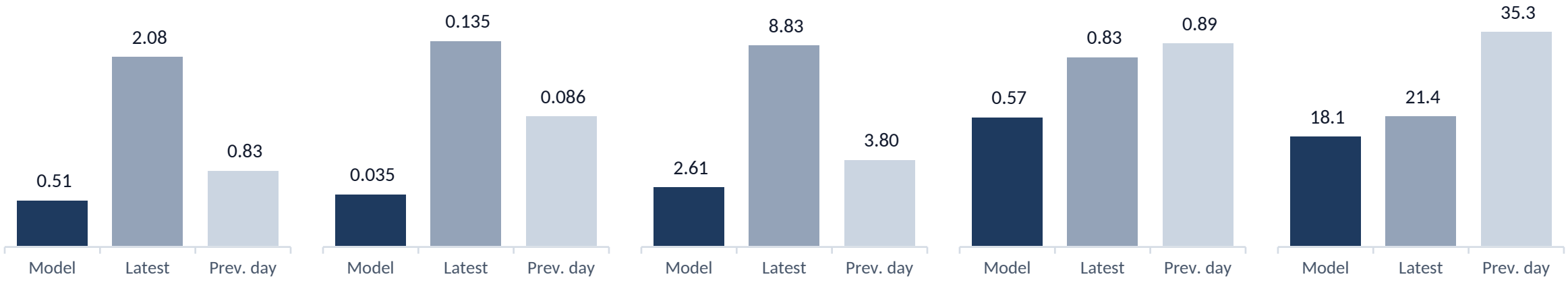
−32%

vs best baseline

Direction
MAE · degrees

−16%

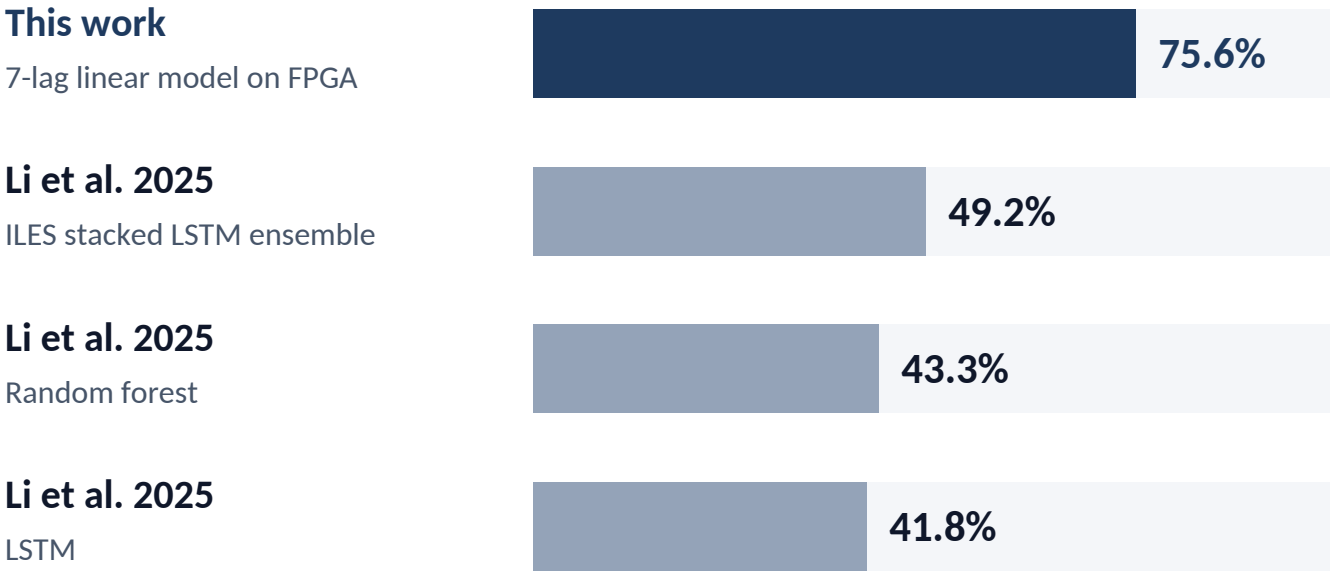
vs best baseline



2025 test set · 8,757 scalar targets · direction: 8,593 non-calm targets (≥ 0.5 m/s), ideal decoding of quantized components

Against the same persistence baseline at 3 hours

MAE reduction versus persistence, 3-hour horizon



	This work	Li et al.
Target	2 m air temp.	Road surface
Site	Ho Chi Minh City	Jiangsu, China
Test data	2025, all year	Winters 2020–24
Persistence	2.08 °C	2.50 °C
Best model	0.51 °C	1.27 °C
Runs on	FPGA, 18-bit	Software, float

$$\text{reduction} = 1 - \text{MAE}_{\text{model}} / \text{MAE}_{\text{persistence}}$$

Different site, variable and season: each model is scored only against its own persistence baseline.

Source: Li et al. 2025, EGUsphere, doi:10.5194/egusphere-2025-3638, Table 5

The design fits in a fifth of the Cyclone II

6,253

logic elements

19% of EP2C35F672C6

6

18 × 18 multipliers

12 of 70 nine-bit elements

50 MHz

physical test clock

+7.459 ns setup slack

79.74 MHz

fitted Fmax estimate

not a measured clock sweep

Flow Summary	
Flow Status	Successful - Tue Oct 27 20:50:14 2026
Quartus II 32-bit Version	13.0.1 Build 232 06/12/2013 SP 1 SJ Web Edition
Revision Name	weather_jtag
Top-level Entity Name	de2_weather_jtag
Family	Cyclone II
Device	EP2C35F672C6
Timing Models	Final
Total logic elements	6,253 / 33,216 (19 %)
Total combinational functions	4,147 / 33,216 (12 %)
Dedicated logic registers	4,571 / 33,216 (14 %)
Total registers	4571
Total pins	2 / 475 (< 1 %)
Total virtual pins	0
Total memory bits	0 / 483,840 (0 %)
Embedded Multiplier 9-bit elements	12 / 70 (17 %)
Total PLLs	0 / 4 (0 %)

Native Quartus II 13.0 Flow Summary · uninstrumented JTAG build · no values altered

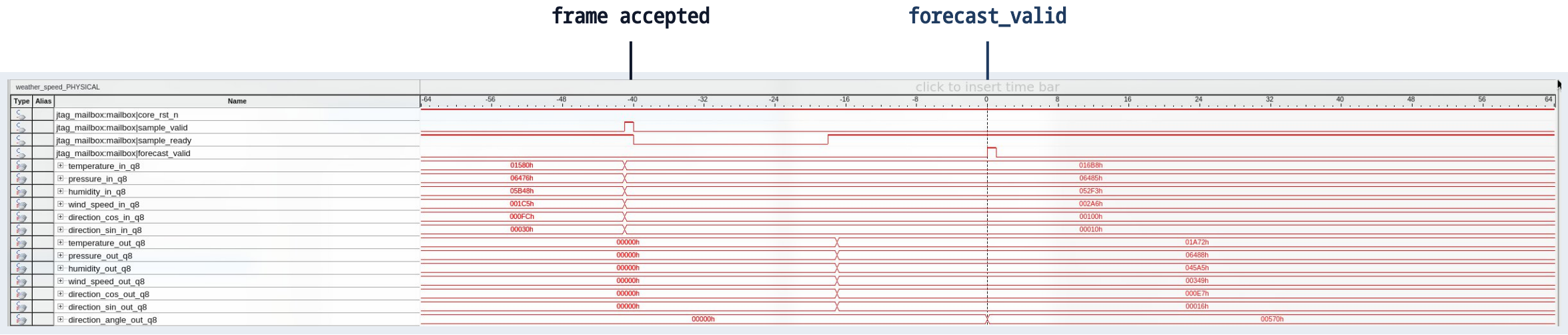
Fit includes the board JTAG wrapper and excludes SignalTap. Timing panels in backup B5.

Published FPGA predictors: different devices, same idea

Design	Task	Device · arithmetic	Resources	Timing
This work CE436 · 2026	**5 weather quantities, 6 lanes × 7 taps**	**Cyclone II EP2C35** 18-bit fixed point	**6,253 LEs** 12 × 9-bit multipliers	**0.80 µs · 40 cycles** RTL latency, 50 MHz board clock
Vásquez et al. Cogn. Comput. 2013	Temperature, MLP network	Xilinx fixed point	not reported in abstract	74 ns max. delay abstract-reported
Ferreira et al. SBCCI 2019	Linear regression fit + 8 predictions	Cyclone IV EP4CE115 32-bit fixed point	8% of 114,480 LEs 89% of 532 multipliers	“one clock cycle” no Fmax reported
Silva et al. Sensors 2022	Rolling linear-regression predictor	Virtex-6 32-bit float	3,440 LUTs 9 multipliers	40.25 ns processing synthesis result
Royer et al. DCIS 2011	Block linear-regression fit	Xilinx DSP48E fixed point	66 slices 5 DSP48E	458 MHz Fmax estimate

LEs, LUTs and slices are different units, and timing evidence ranges from board tests to estimates: context, not a ranking.

A forecast captured on the physical board



Native SignalTap capture, separately instrumented build · STP and decoded CSV retained

104

forecasts checked
from 128 input frames

`temperature_out_q8`

`01A72h → 6770 / 256 → 26.4453 °C`

41 acquisition intervals ≈ documented 40-cycle RTL latency (sampling convention)

What we demonstrated, and what we did not claim

Demonstrated

- ✓ Five quantities forecast on the FPGA
- ✓ Recorded observations replayed over JTAG
- ✓ Fixed, pre-trained coefficients
- ✓ Physical replay at 50 MHz

Further work

- ○ Live sensor acquisition
- ○ Other sites and years
- ○ Rainfall probability
- ○ Power measurement and clock sweep

Hourly forecasting does not need sub-microsecond speed: the contribution is a verified hardware implementation and its schedule.

CE436 · Final defense

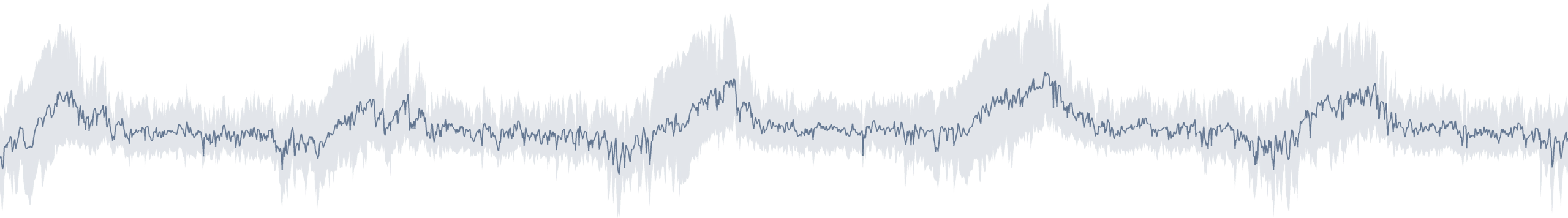
Weather inference, verified on FPGA

5
forecast quantities

0.80 µs
RTL latency at 50 MHz

104
physical forecasts checked

Thank you. Questions are welcome.



Backup B1

The original temperature model: a seasonal anchor and two corrections

$$\hat{T}(t+3) = T(t-21) + a \cdot \Delta daily + b \cdot \Delta recent + c$$

```
Δdaily  = T(t) - T(t - 24)
Δrecent = T(t) - T(t - 3)
```

Parameter	Fitted value
a	0.7454870439
b	0.0091494450
c	0.0011295817 °C

Temperature test MAE **0.538017 °C** prototype → **0.507906 °C** current model

Backup B2

The temperature lane: seven deployed weights and a Q8 bias

Lag (h)	Input Q8	Weight Q14	Product
0	7680	20035	153868800
1	7552	−9335	−70497920
3	7168	1621	11619328
6	6912	−205	−1416960
12	6656	15	99840
21	7936	13908	110373888
24	7424	−10382	−77075968

Sum including aligned bias

$$S = 132050048$$

Scale back to Q8

$$\lfloor S / 16384 \rfloor = 8059$$

8059 / 256 = 31.48 °C

Backup B3

Test errors keep the physical unit of each quantity

Target	MAE	RMSE
Temperature (°C)	0.507906	0.730939
Pressure (kPa)	0.034600	0.044638
Humidity (percentage points)	2.614292	3.685767
Wind speed (m/s)	0.566878	0.750218
Direction (degrees)	18.084598	30.201056

8,757 scalar targets. Direction: 8,593 targets with wind speed ≥ 0.5 m/s, ideal atan2 of quantized components. CORDIC maximum conversion error: 0.005635°.

Backup B4

Each lane repeats operand, product and accumulate for seven terms

Edge	Operation
`N`	Accept frame, shift history, initialize bias
`N + 1`	Capture lag-0 operand
`N + 2`	Capture product
`N + 3`	Add product to accumulator
`N + 4 … N + 21`	Repeat for the remaining six terms
`N + 22`	Scale, limit and publish lane result
`N + 23`	Earliest next eligible acceptance

Synchronous active-low reset cancels pending work and restarts history eligibility.

Backup B5

Native TimeQuest panels record the fitted timing margins

Multicorner Timing Analysis Summary

	Clock	Setup	Hold	Recovery	Removal	Minimum Pulse Width
1	Worst-case Slack	7.459	0.215	N/A	N/A	7.981
1	CLOCK_50	7.459	0.215	N/A	N/A	7.981
2	altera_reserved_tck	N/A	N/A	N/A	N/A	97.778
2	Design-wide TNS	0.0	0.0	0.0	0.0	0.0
1	CLOCK_50	0.000	0.000	N/A	N/A	0.000
2	altera_reserved_tck	N/A	N/A	N/A	N/A	0.000

Slow Model Fmax Summary

	Fmax	Restricted Fmax	Clock Name	Note
1	79.74 MHz	79.74 MHz	CLOCK_50	

Physical test clock

50 MHz

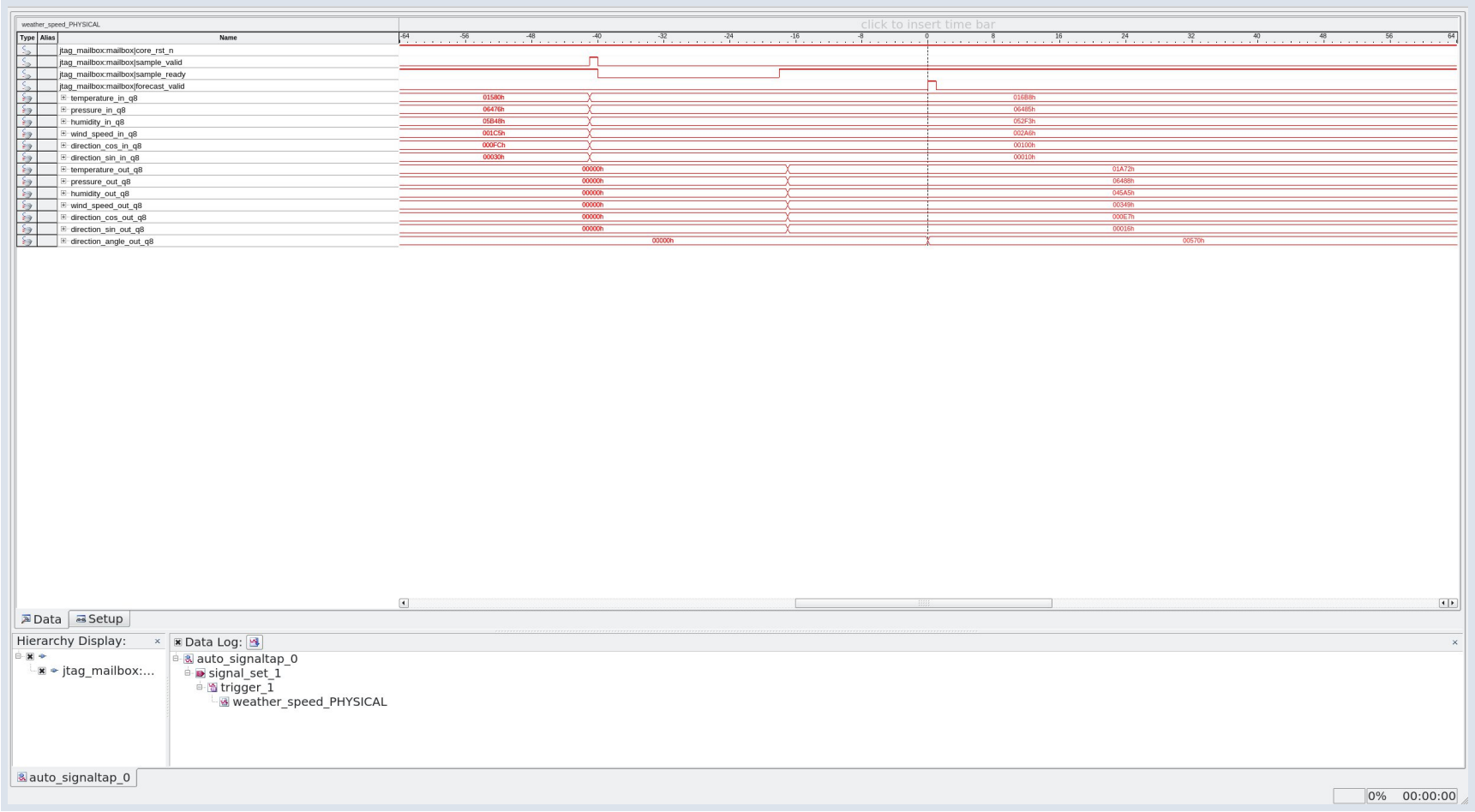
Fitted estimate

79.74 MHz

Setup slack 7.459 ns · hold 0.215 ns

Backup B6

The original SignalTap acquisition with decoded buses and trigger



Backup B7

The JTAG mailbox holds each request and keeps the completed forecast

01 **Stage payload** → 02 **Toggle request** → 03 **Wait for acknowledgement** → 04 **Read stable result**

128-bit
source word (host → FPGA)

256-bit
probe word (FPGA → host)

1
outstanding request at a time

Warm-up replies acknowledge without a forecast.

Backup B8

Source records behind the model, implementation and physical results

Model and evaluation

```
versions/weather_v2/models/model.json
versions/weather_v2/reports/metrics.csv
versions/weather_v2/data/weather.csv
```

Implementation

```
docs/reports/JTAG_IMPLEMENTATION_REPORT.md
docs/reports/weather_speed_modelsim_timing.json
```

Physical session

```
docs/reports/defense_session/
Native screenshots, replay logs, STP and decoded CSV
```

Related work

```
Li et al. 2025       doi:10.5194/egusphere-2025-3638
Vásquez et al. 2013  doi:10.1007/s12559-012-9158-z
Ferreira et al. 2019 doi:10.1145/3338852.3339853
Silva et al. 2022    doi:10.3390/s22093556
Royer et al. 2011    DCIS · docs/specs/ols_fpga_literature_review.md
```